

ChatGPT Student Survey Analysis

This repository contains the data and code used to explore student perceptions of ChatGPT in education, skills development, and the workforce.

****Original Dataset Source**** :<https://data.mendeley.com/datasets/ymg9nsn6kn/2>

📖 Further Reading

For a more detailed analysis and discussion, check out our Medium articles:

- 📖 [Part 1: What Students Really Think About ChatGPT: Impact on Study Difficulty, Emotions](<https://medium.com/@arabellyabhinav28/what-students-really-think-about-chatgpt-a-data-driven-analysis-of-chatgpts-impact-in-education-2a13da7f64c9>)
- 📖 [Part 2: ChatGPT's Impact on Employment, Access, and Challenging Fields](<https://medium.com/@yashchaudhari33/3dd999360201>)

📁 Repository Structure

1. `Data/`

- ****Cleaned Dataset****: Scripts to preprocess and clean the original survey dataset.
- ****Questionnaire****: Survey questions used to collect responses.

2. `Final Dataset/`

- ChatGPT_Education_Analysis.csv - Final Dataset - Ready to use for Analysis

3. `Code/`

- ****Part 1: Data Analysis****:

Code for initial data exploration, summary statistics. We further explore:

- Students' emotional responses to using ChatGPT.
- Differences in ChatGPT usage extent across age groups.
- Comparison between free and paid ChatGPT users.
- Patterns in how students use ChatGPT for various academic tasks.

- ****Part 2: Data Analysis****:

🛠️ Technologies Used

- Python (Pandas, Matplotlib, Seaborn)
- Jupyter Notebook
- Git for version control

📊 Project Overview

We analyzed thousands of student survey responses to understand:

- How students use ChatGPT across different academic difficulty levels ?
- The perceived impact of AI tools on creativity, knowledge, Employment and career preparation
- Differences in ChatGPT usage across age groups, education levels, and student status

📧 Contact

For any questions or suggestions, feel free to reach out

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Yash Chaudhari!

What Students Really Think About ChatGPT: A Data-Driven Analysis of ChatGPT's Impact in Education — Part 1

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[Abhinav Arabelly](#)

4 min read

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Apr 24, 2025

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In a world where AI tools are becoming part of everyday learning, it's important to understand how students actually experience them. Not just how often they use these tools — but how they shape learning outcomes, motivation, and confidence in real academic settings.

This analysis is based on the [Global ChatGPT Student Survey](#), conducted by the University of Ljubljana in collaboration with over 200 academic partners worldwide. The dataset includes responses from thousands of higher education students across diverse countries and study levels, capturing how they use ChatGPT, what they feel about it, and how it impacts their academic and personal development.

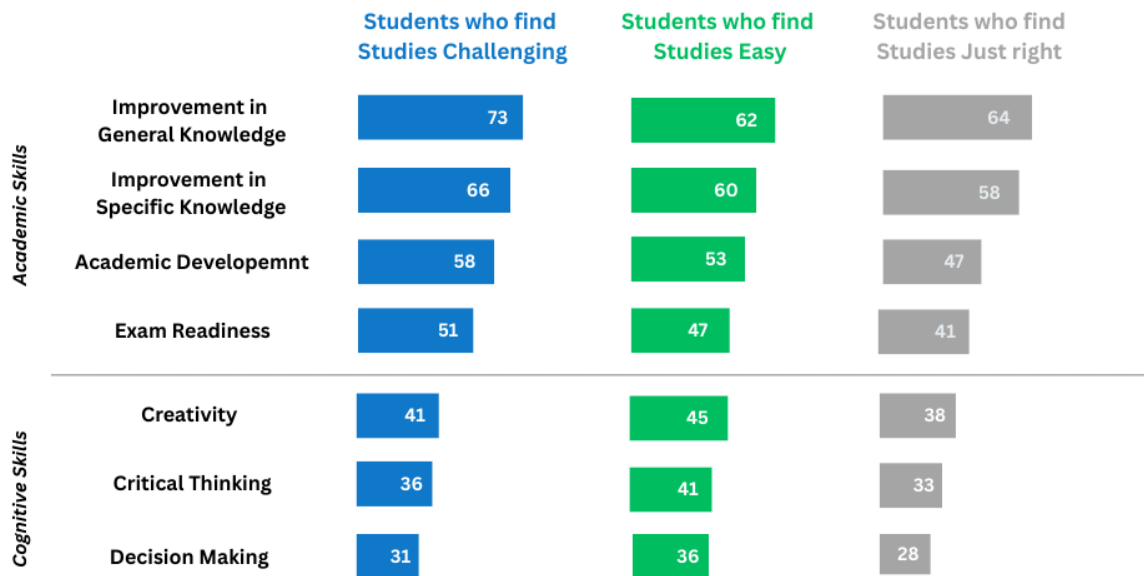
We analysed the data closely and found some fascinating insights worth sharing — on how ChatGPT shapes students' academic journeys, influences their emotions, and supports their personal growth. Let's dive in!

How ChatGPT Supports Students' Skill Development Based on Perceived Study Difficulty ?

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ChatGPT's Role in College Students' Skills Development for Varying Levels of Study Difficulty

% of students who reported being benefited from ChatGPT in different skills, based on how difficult they find their studies?



Students who find studies challenging tend to benefit more in academic areas, while those who find studies easy report greater improvements in cognitive skills

Key Insights:

- **Students who find school *hard* gain more in academic skills.** 73% of them say ChatGPT boosted their general knowledge — compared to just 62% of students who find school easy. Similar trends show up across other academic skills too — suggesting that ChatGPT is more effectively helping these students fill knowledge gaps.

- **Students who find school *easy* are seeing more growth in creativity, critical thinking, and decision-making** than those who struggle. ChatGPT seems to boost higher-order thinking most in students who already feel confident in their learning.
- While ChatGPT helps students go beyond academics and sharpen higher-level thinking skills, **the biggest reported gains are still in academic areas.**

ChatGPT currently plays a stronger role in boosting academic performance than in enhancing cognitive skills.

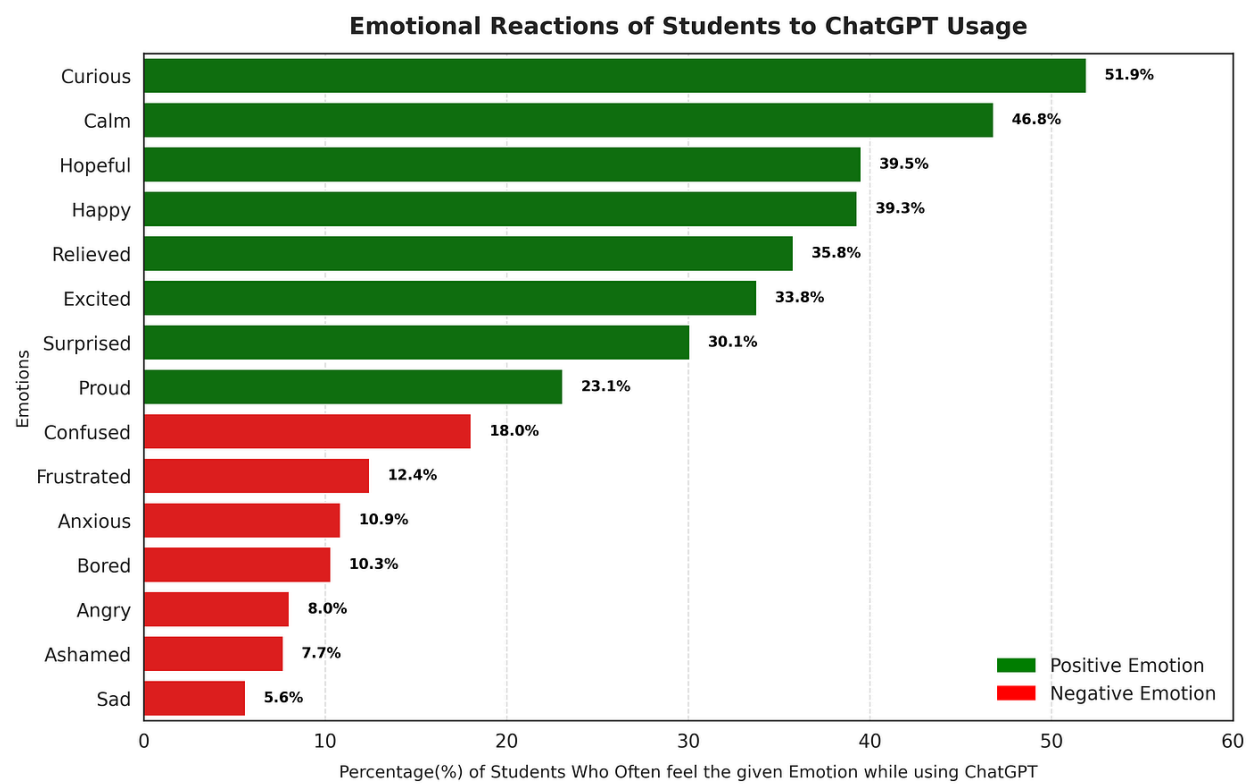
Conclusion and Future outcomes:

These findings **highlight the growing need for personalised learning** — showing that if tools like ChatGPT can adapt to each student's goals, confidence, and challenges in

real time, they could significantly accelerate both academic success and holistic skill development

What Emotions do Student’s experience while using ChatGPT ?

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Key Insights:

- **Most students report feeling positive emotions when using ChatGPT.** Over half of students (51.9%) say ChatGPT makes them feel *curious*, followed by *calm* (46.8%), *hopeful* (39.5%), and *happy* (39.3%). These emotions show that for many, ChatGPT creates a supportive, encouraging learning space.
- **Negative Emotions Exist but are less common.** Around 18% of students report feeling confused, and 12.4% feel frustrated. While less frequent, emotions like anxiety (10.9%), boredom (10.3%), and anger (8%) suggest occasional friction in the user experience, possibly arising from unmet expectations, unclear responses or task complexity.

Conclusion and Future outcomes:

These findings show that ChatGPT plays a positive role in students' emotional learning experience. Although, there's room

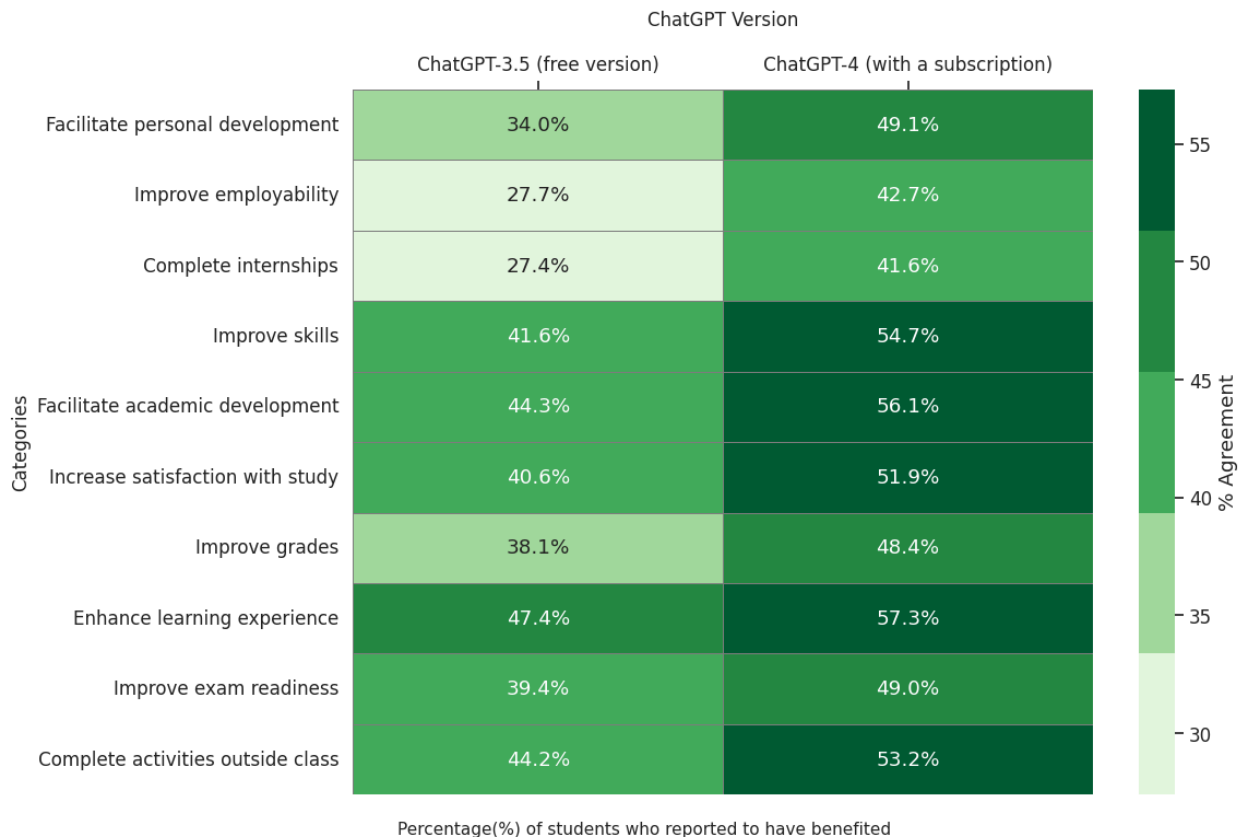
to make the experience even more supportive and responsive.

Looking ahead, there's an opportunity to make **AI tools more emotionally aware** — able to notice when a student feels confused or anxious, and respond with encouragement or a clearer explanation. By blending **emotional intelligence with academic guidance**, the next generation of AI could become not just a tutor — but a **truly empathetic learning companion**.

ChatGPT-4 (Subscription) v/s 3.5 (Free): Do Students Really Benefit More from Subscription model?

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ChatGPT-4 (Subscription) vs 3.5 (Free): Do Students Really Benefit More?



Key Insights:

- **Students using ChatGPT-4 (Subscription) report significantly greater benefits in key academic and career areas.** From improving employability (42.7% vs 27.7%) to completing internships (41.6% vs 27.4%), enhancing skills (54.7% vs 41.6%), and even

boosting grades (48.4% vs 38.1%), the paid version consistently outperforms the free version in supporting students' academic development.

- The **advantages go beyond academics — supporting broader personal growth as well.**

Subscription users were more likely to say ChatGPT helped them facilitate personal development (49.1% vs 34.0%) and complete activities outside of class (53.2% vs 44.2%), suggesting that ChatGPT plays a role in helping students stay motivated, organised, and engaged beyond the classroom.

Conclusion and Future outcomes:

It was interesting to see that the subscription-based ChatGPT-4 had a significantly stronger impact on students, especially in career and personal development. Making the subscription based models more **affordable or accessible to all students** could

help close opportunity gaps and ensure that a wider range of learners benefit from AI-enhanced education.

This piece was co-written with [Yash Chaudhari](#) as Part 1 of a shared series on student perceptions of ChatGPT.

👉 Don't miss [Part 2](#) on their profile, which explores how students expect ChatGPT to shape the job market, whether access to it remains a privilege, and its growing use among those in more demanding academic fields.

Ai In Education

ChatGPT

Data Analysis

What Students Really Think About ChatGPT: A Data-Driven

Analysis of ChatGPT's Impact in Education — Part 2



Yash Chaudhari

Following

3 min read

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In [Part 1](#) of our analysis, we explored how students are using ChatGPT to navigate academic challenges, build confidence, and shape their learning experiences. We looked at emotional engagement, skill development, and how different versions of ChatGPT (free vs. subscription) are impacting outcomes.

But there's more to this story.

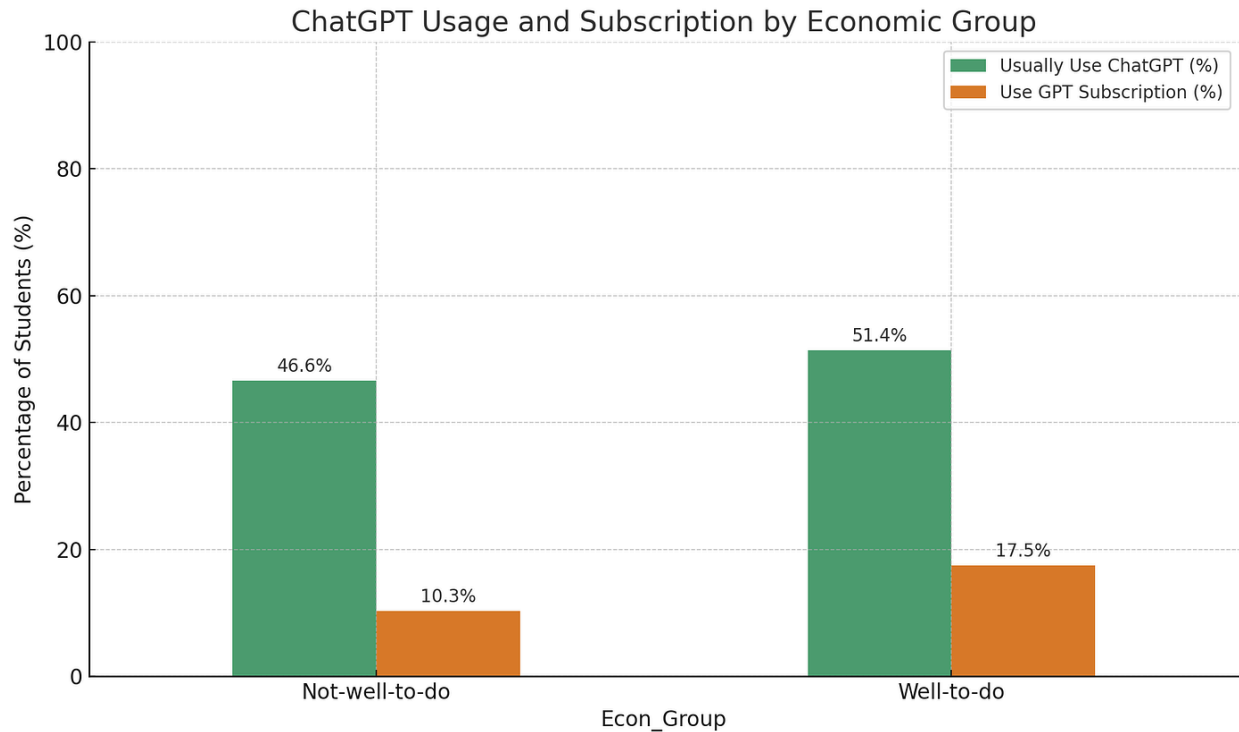
In Part 2, we shift focus to how students **perceive ChatGPT in the broader context of education, equity, and employability**. Who really has access to premium AI tools? How does ChatGPT support students in more demanding fields? Can it actually bridge the skills gap? And are students worried it might take their jobs?

We unpack all of this using data from the [Global ChatGPT Student Survey](#), run by the University of Ljubljana in collaboration with over 200 academic institutions worldwide. The results? Honest, sometimes surprising, and full of insight.

Let's dive in.

Is Access to AI Tools Still a Privilege?

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Students from wealthier backgrounds are not only more likely to use ChatGPT, but are nearly twice as likely to subscribe to premium versions. The data reveals a clear link between economic status and access to advanced AI features.

Key Insights:

- Students from **higher economic backgrounds** are significantly more likely to use ChatGPT — and pay for it. Over **51% of well-to-do students** regularly use ChatGPT, and **17.5% subscribe to GPT-4 or use both versions**, compared to just **10.3% among less well-off peers**.

- The **subscription gap matters** — GPT-4 users report better outcomes in career prep and skill development.

Conclusion and Future Outcomes:

While ChatGPT opens doors for many, premium features are still gated. Making AI tools more financially accessible could play a major role in closing opportunity gaps in digital education.

Challenging Fields? ChatGPT Might Be the Real MVP

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In more academically demanding fields like Applied Sciences and Natural Sciences, students report both higher study difficulty and higher ChatGPT usage suggesting AI is becoming a tool to cope with complexity

Key Insights:

- Fields like **Applied Sciences** not only report **more academic difficulty**, but also show **higher ChatGPT usage**.

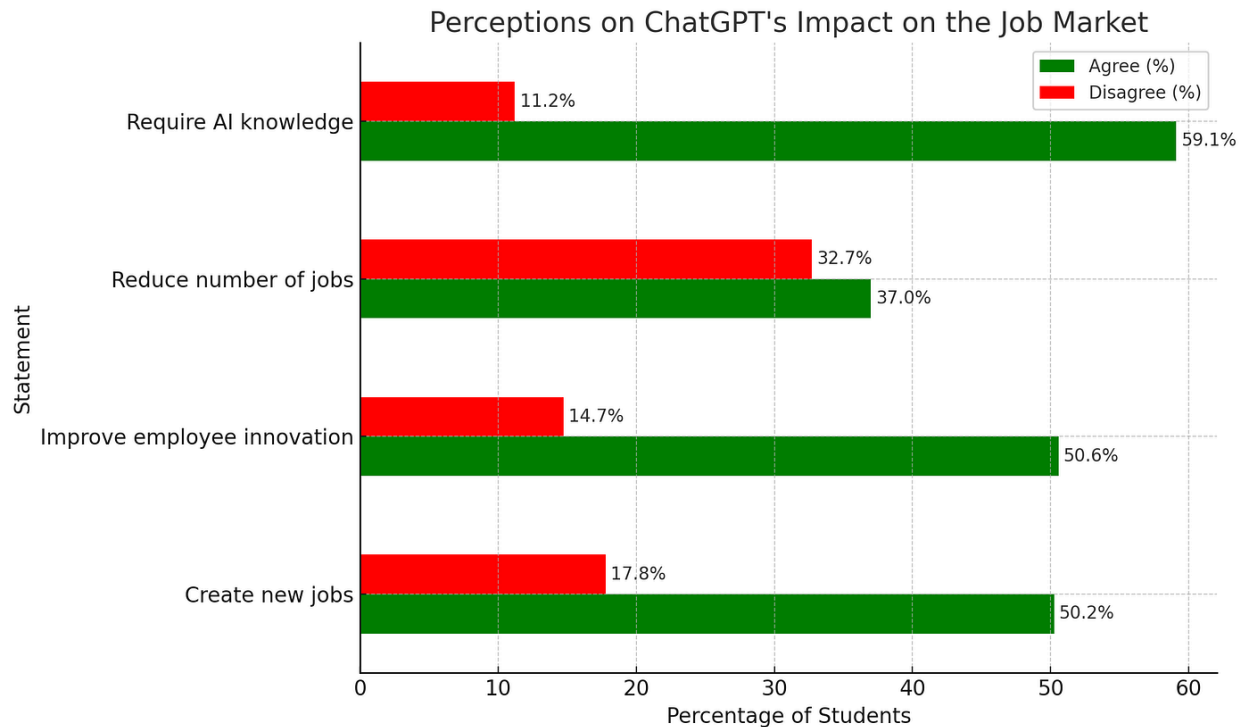
- Meanwhile, students in **Arts and Humanities** report lower use and less perceived difficulty — a subtle but telling contrast.

Conclusion and Future Outcomes:

Students facing tougher coursework are leaning on ChatGPT as a cognitive co-pilot — not a crutch. This suggests AI can serve as a personalized support system in rigorous academic fields, helping students not just keep up, but stay motivated.

Students Expect AI to Reshape the Job Market — For Better and Worse

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While students recognize that AI may displace some jobs, even more believe it will create new ones, demand AI-related skills, and foster workplace innovation, signaling a readiness for an evolving job landscape.

Key Insights:

- While **37% of students believe** ChatGPT could reduce jobs, a greater number believe it will also **create new roles (50%)**, **enhance innovation (51%)**, and **require basic AI tools knowledge (59%)** in future jobs.

Conclusion and Future Outcomes:

This isn't fear — it's foresight. Students know the nature of work is evolving, and they're preparing to adapt. Educational systems must evolve too — helping students build the confidence and flexibility to thrive in AI-augmented careers.

Final Thoughts

These insights collectively show one thing: **students aren't resisting AI — they're embracing it, questioning it, and integrating it** into how they learn, grow, and prepare for the future.

If institutions, educators, and tech leaders want to meet students where they are, we need to build AI tools that are not only powerful — but equitable, ethical, and emotionally intelligent.

This piece was co-written with [Abhinav Arabelly](#) as Part 2 of a shared series on student perceptions of ChatGPT.

👉 Don't miss [Part 1 on their profile](#), which explores ChatGPT's impact on academic outcomes and student emotions.